For a norovirus-infected adult, what is the total viral load in a single projectile vomiting episode, and how does it deposit on nearby surfaces? Give: emesis volume (mL), norovirus genome copies per mL of vomitus, the surface area over which droplets deposit, and the fraction of the emitted load that lands on surfaces versus becoming aerosol. Include measured values with citations and note which are estimates.

Norovirus Vomiting: Viral Load, Surface Deposition, and Aerosol Fraction

A single projectile vomiting episode from a norovirus-infected adult can emit **~10^5–10^9 viral genome copies** across an estimated 50–1,350 mL of vomitus, with the vast majority depositing on surfaces over ~7.8 m² and only a tiny fraction (<0.03%) becoming aerosol (Kirby et al., 2016; Tung-Thompson et al., 2015; Booth, 2014).

Key Quantitative Parameters

Parameter	Value	Type	Source
Emesis volume per episode	50–800 mL (estimated max); 0.4–1.35 L (iatrogenic)	**Estimate**	(Tung-Thompson et al., 2015; Booth, 2014; Booth & Frost, 2019)
Norovirus titer in vomitus (GI)	8.0×10^5 GEC/mL (mean)	**Measured**	(Kirby et al., 2016)
Norovirus titer in vomitus (GII)	3.9×10^4 GEC/mL (mean)	**Measured**	(Kirby et al., 2016)
Median titer (Norwalk virus)	4.1×10^4 GEq/mL	**Measured**	(Atmar et al., 2014; Hagbom et al., 2021)
Cumulative shedding per subject	1.8×10^8 GEC (all episodes)	**Measured**	(Kirby et al., 2016)
Total vomit shedding (dose-response)	6.4×10^5 – 3.0×10^7 GEC	**Measured**	(Ge et al., 2023)
Surface deposition area	≥7.8 m² (forward >3 m, lateral 2.6 m)	**Measured (simulated)**	(Booth, 2014; Kirby et al., 2016)
Aerosolized fraction of total virus	7.2×10^{-5}% – 2.67×10^{-2}%	**Measured (surrogate)**	(Tung-Thompson et al., 2015)

Parameter	Value	Type	Source
Airborne NoV concentration post-vomiting	5–215 copies/m³	**Measured (outbreak)**	(Alsved et al., 2019; Bonifait et al., 2015)

FIGURE 1 Measured and estimated parameters for norovirus vomiting episodes

Viral Load in Vomitus

Human challenge studies provide the only directly measured vomitus titers. GI.1 Norwalk virus yielded a mean of **8.0 × 10⁵ GEC/mL**, while GII strains averaged 3.9 × 10⁴ GEC/mL, a statistically significant difference ($p = 0.02$) (Kirby et al., 2016). A separate Norwalk virus challenge found virus in 56% of vomitus samples at a **median concentration of 4.1 × 10⁴ GEq/mL** (Atmar et al., 2014). Cumulative shedding across all vomiting episodes per subject reached 1.8 × 10⁸ GEC for Norwalk and Snow Mountain viruses combined (Kirby et al., 2016).

Detection depended on episode frequency: subjects who vomited only once never had detectable virus in emesis, whereas 57% of first samples from subjects with **multiple episodes** were positive (Kirby et al., 2016). Viral titers tended to **increase with each successive** vomiting event (Kirby et al., 2016). A dose-escalation challenge study showed total vomit shedding rising from 6.4 × 10⁵ to 3.0 × 10⁷ GEC as inoculum increased from 4.8 to 4,800 RT-PCR units (Ge et al., 2023).

Early electron microscopy estimated **~3 × 10⁷ particles in a 30-mL bolus** of vomit, implying ~10⁹ particles/L (Booth & Frost, 2019). This figure is an estimate, and a recent systematic review notes that quantitative vomitus shedding data remain scarce compared with stool data (Zheng et al., 2025).

Surface Deposition and Aerosol Fraction

Simulated vomiting experiments using the "Vomiting Larry" system demonstrated droplet spread exceeding **3 m forward and 2.6 m lateral**, covering an area of at least 7.8 m² (Booth, 2014). Viable surrogate virus (feline calicivirus) was recovered from 88 of 90 floor samples at concentrations ≥10 PFU/mL, including droplets at the **3-meter maximum distance** (Booth & Frost, 2019). Recovery controls showed swabs captured only ~0.3% and sponges ~4% of applied virus, meaning actual surface contamination is likely far higher than measured (Booth & Frost, 2019).

The aerosolized fraction was quantified in a similitude-based model using bacteriophage MS2: across all conditions, **<0.03% of total virus** was aerosolized, ranging from 7.2×10^{-5}% to 2.67×10^{-2}% (Tung-Thompson et al., 2015). Despite this tiny fraction, the absolute numbers (36–13,350 PFU) exceed the norovirus **infectious dose of 18–1,300 particles** (Tung-Thompson et al., 2015; Tan et al., 2024; Rupprom et al., 2024).

- Airborne norovirus RNA was detected at 5–215 copies/m³ in hospital outbreak rooms, strongly associated with vomiting within the prior 3 hours (OR 8.1) (Alsved et al., 2019).
- Norovirus-containing particles <1 µm were detected, with a **10–30% deposition probability** in the mouth, nose, or airways upon inhalation (Alsved et al., 2019; Tan et al., 2024).
- Vomitus can contain ~10⁷ genome copies/mL, and each vomiting event has been estimated to carry the potential to infect >150,000 susceptible persons (Purhonen et al., 2024; Hagbom et al., 2021).

FIGURE 2 Surface vs. aerosol deposition of norovirus from vomiting

Evidence Strength and Limitations

No study has simultaneously measured all four requested parameters in a single norovirus-infected human. Emesis volume derives from expert opinion and ipecacuanha-induced emesis in non-norovirus patients (Tung-Thompson et al., 2015; Booth & Frost, 2019). Viral titers come from human challenge studies with small sample sizes (Kirby et al., 2016; Atmar et al., 2014). Surface deposition area and aerosol fraction rely on surrogate viruses (MS2, FCV) in physical models, not wild-type human norovirus (Tung-Thompson et al., 2015; Booth & Frost, 2019). Airborne concentrations are from outbreak investigations that cannot isolate vomiting from other sources like toilet flushing or fomite resuspension (Bonifait et al., 2015; Alsved et al., 2019). Norovirus in vomitus has been confirmed infectious via replication in human intestinal enteroids, validating that detected RNA can represent viable virus (Hagbom et al., 2021).

These values are research syntheses, not clinical guidance. Infection control decisions should follow institutional protocols and public health guidance.

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